# Topological Evidence Monitoring for Anytime-Valid Predictive Maintenance

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**Abstract**

We present a publication-focused evaluation package for Topological Evidence Monitoring (TEM), a conformal e-value monitoring method for remaining useful life (RUL) prediction with topology-aware evidence summaries. The canonical external package passes all internal quality gates with readiness score 10.0/10 and conservative readiness score 10.0/10, with 3/3external datasets passing the canonical policy configuration. We provide a full artifact-linked manuscript build, theorem statements aligned to fixed-grid e-process monitoring, and policy replay experiments from frozen checkpoints/calibration bundles to quantify the validity-sharpness tradeoff.

## 1 Introduction

Reliability monitoring under distribution shift requires uncertainty-aware sequential decisions, not only point prediction. Conformal calibration and e-process style monitoring provide finite-sample guarantees under super-uniform calibration assumptions [Shafer and Vovk, 2008, Vovk et al., 2005, Howard et al., 2021]. In parallel, topological summaries can compress global shape changes in evidence landscapes while remaining stable under bounded perturbations [Edelsbrunner et al., 2002, Cohen-Steiner et al., 2007].

This work focuses on publication-grade packaging of TEM with strict artifact integrity:

- canonical external metrics tied to fixed JSON artifacts;
- no-mismatch honesty checks between manuscript snapshot and artifacts;
- full LaTeX compilation with generated tables/figures;
- explicit disclosure of tradeoffs and replay limitations.

## 2 Method Summary

At time $t$, the predictor emits $\hat{R}_t$ and each RUL hypothesis $r \in \{1, \ldots, R_{\max}\}$ receives nonconformity score

$$s_t(r) = |\hat{R}_t - r|, \quad p_t(r) = \frac{1 + \#\{i : A_i^{cal} \geq s_t(r)\}}{n_{cal} + 1}. \tag{1}$$

Evidence is updated by

$$K_t(r) = K_{t-1}(r)\left(1 + \lambda_t(1 - 2p_t(r))\right), \quad K_0(r) = 1, \tag{2}$$

Table 1: Canonical external results (FD001 policy v8).

| Dataset | RMSE | MAE | $\text{RMSE}_{last}$ | $\text{MAE}_{last}$ | RUL cov. | Tau viol. |
|---|---|---|---|---|---|---|
| femto | 14.362 | 7.993 | 52.873 | 50.395 | 1.000 | 0.000 |
| xjtu_sy | 30.133 | 23.684 | 33.171 | 33.156 | 1.000 | 0.000 |
| cmapss | 16.396 | 10.992 | 16.623 | 12.219 | 1.000 | 0.000 |

with log evidence $V_t(r) = \log K_t(r)$ and confidence set

$$C_t^\alpha = \{r : K_t(r) < 1/\alpha\}. \tag{3}$$

Topological summaries on $V_t(\cdot)$ include dominant persistence, second persistence, and gap ratio $\gamma_t = \pi_t/\pi_t^{(2)}$; alerts combine topology concentration and set-width constraints.

## 3 Theoretical Statements

**Theorem 1** (Anytime validity, fixed-grid inversion)**.** *Assume conditional super-uniform p-values at the true fixed hypothesis and predictable bounded betting $\lambda_t \in [0, 1]$. Then $K_t(h^\star)$ is a nonnegative supermartingale and*

$$\Pr\left(\sup_{t\geq 0} K_t(h^\star) \geq 1/\alpha\right) \leq \alpha.$$

*Hence* $\Pr(\exists t : h^\star \notin C_t^\alpha) \leq \alpha$.

**Theorem 2** (Persistence stability)**.** *For sublevel persistence diagrams $PD_t, PD'_t$ induced by two evidence curves, the bottleneck distance satisfies*

$$d_B(PD_t, PD'_t) \leq \|V_t - V'_t\|_\infty.$$

*Therefore Lipschitz functionals of persistence (e.g., top lifetime and clipped gap ratio) are stable under bounded predictor perturbations.*

**Remark 1.** *Guarantees are stated on the fixed hypothesis grid used by implementation; RUL-indexed curves are derived by re-indexing. This avoids mixing fixed-parameter validity and moving-index interpretation.*

## 4 Experimental Protocol

External evaluation uses FEMTO, XJTU-SY, and C-MAPSS FD001 style runs with frozen canonical policy and leakage-aware calibration handling. Internal comparator packaging includes strict-main and split-conformal variants across FD001–FD004.

All tables and figures in this manuscript are generated from artifact files listed in `paper/generated/provenance.json`.

Table 2: Policy replay from fixed checkpoints/calibration bundles. RMSE is unchanged by policy replay; coverage and tau violation are policy-sensitive.

| Policy | Dataset | $\alpha$ | $\lambda$ | margin | RUL cov. | Tau viol. | RMSE |
|---|---|---|---|---|---|---|---|
| Canonical | femto | 0.001 | 0.030 | 0.300 | 1.000 | 0.000 | 14.362 |
| Canonical | xjtu_sy | 0.001 | 0.030 | 0.300 | 1.000 | 0.000 | 30.133 |
| Canonical | cmapss | 0.001 | 0.030 | 0.300 | 1.000 | 0.000 | 16.396 |
| Balanced | femto | 0.002 | 0.030 | 0.150 | 0.316 | 1.000 | 14.362 |
| Balanced | xjtu_sy | 0.002 | 0.030 | 0.150 | 1.000 | 0.000 | 30.133 |
| Balanced | cmapss | 0.002 | 0.030 | 0.150 | 1.000 | 0.000 | 16.396 |
| Aggressive | femto | 0.005 | 0.050 | 0.050 | 0.092 | 1.000 | 14.362 |
| Aggressive | xjtu_sy | 0.005 | 0.050 | 0.050 | 1.000 | 0.000 | 30.133 |
| Aggressive | cmapss | 0.005 | 0.050 | 0.050 | 1.000 | 0.000 | 16.396 |
| Robust | femto | 0.003 | 0.040 | 0.300 | 1.000 | 0.000 | 14.362 |
| Robust | xjtu_sy | 0.003 | 0.040 | 0.300 | 1.000 | 0.000 | 30.133 |
| Robust | cmapss | 0.003 | 0.040 | 0.300 | 1.000 | 0.000 | 16.396 |

Table 3: True retrain robustness check (no checkpoint reuse).

| Dataset | RMSE | RUL cov. | Tau viol. | Runs |
|---|---|---|---|---|
| femto | 15.243 | 1.000 | 0.000 | 4 |

# 5 Results

## 5.1 Canonical External and Internal Readiness

## 5.2 Policy Replay Tradeoff

Policy replay keeps model weights and calibration bundles fixed and only changes $(\alpha, \lambda, \text{margin})$. As expected, RMSE remains invariant while validity metrics can shift substantially. In our replay runs, FEMTO validity degrades sharply under less conservative settings, while XJTU-SY and C-MAPSS remain stable for the same parameter changes. To tighten this analysis beyond three hand-picked points, we run replay-only sweeps from frozen artifacts and include the strongest cross-dataset validity-preserving candidate as a robust replay row. In the all-dataset sweep, settings with margin 0.30 satisfy the coverage/tau targets, while reducing margin to 0.25 causes immediate FEMTO validity failures. Paired run-level comparator tests are reported with Holm correction, and we include a policy-level sharpness table to make conservatism-vs-validity tradeoffs explicit. We also include a true retrain robustness block (no checkpoint reuse) to validate that conclusions are not purely replay artifacts.

## 5.3 Figures

# 6 Reproducibility and Integrity

The manuscript build is coupled to automated checks:

- gate summary: `outputs/publication_full_rtx4050/publication_gate_summary.json`;

Table 4: Comparator package summary on internal FD001–FD004 matrix.

| Method | RMSE mean | RUL cov. mean | Tau viol. mean |
|---|---|---|---|
| strict_main | 22.459 | 0.988 | 0.047 |
| marginal_rul | 22.459 | 0.910 | 0.000 |
| alpha_0p1 | 22.459 | 0.979 | 0.068 |
| no_margin_matched_bins | 22.459 | 0.959 | 0.152 |
| split_conformal_global_a0p05 | 22.459 | 0.947 | 0.309 |
| split_conformal_global_a0p01 | 22.459 | 0.983 | 0.116 |
| split_conformal_conditional_a0p05 | 22.459 | 0.906 | 0.460 |
| split_conformal_conditional_a0p01 | 22.459 | 0.959 | 0.249 |
| deep_ensemble_gaussian_95 | 19.532 | 0.381 | 0.988 |
| deep_ensemble_gaussian_99 | 19.532 | 0.471 | 0.972 |
| deep_ensemble_conformalized_a0p05 | 19.532 | 0.972 | 0.165 |
| deep_ensemble_conformalized_a0p01 | 19.532 | 0.993 | 0.057 |

Table 5: Paired run-level significance summary (Holm-adjusted).

| Comparator | Metric | Effect (main-comp) | Win rate | $p$ | $p_{Holm}$ |
|---|---|---|---|---|---|
| marginal_rul | coverage | 0.0692 | 0.369 | 0.0000 | 0.0000 |
| no_margin_matched_bins | coverage | 0.0307 | 0.129 | 0.0000 | 0.0000 |
| no_margin_matched_bins | tau_violation | -0.1095 | 0.110 | 0.0000 | 0.0000 |
| alpha_0p1 | coverage | 0.0082 | 0.057 | 0.0000 | 0.0000 |
| alpha_0p1 | tau_violation | -0.0251 | 0.025 | 0.0001 | 0.0001 |

- paper honesty report: `outputs/publication_full_rtx4050/paper_release/reports/paper_honesty_report.json`;
- generated table provenance: `paper/generated/provenance.json`.

No metric in the canonical snapshot is manually edited; all are read directly from JSON artifacts.

## 7 Limitations

- Conservative policy settings can over-prioritize validity margin versus sharpness.
- Replay frontier granularity is still moderate; a denser grid could map the Pareto boundary more precisely.
- True retrain robustness currently covers only a limited external subset; broad retrain coverage remains future work.
- Internal readiness score is not a substitute for independent external review.

## 8 Conclusion

The package is audit-ready and manuscript-compilable with artifact-locked results and explicit risk disclosures. Canonical external claims are reproducible and pass integrity checks. Remaining pub-

Table 6: Policy-level sharpness/validity summary.

| Policy | Cov. mean | Tau max | Width mean | Pareto |
|---|---|---|---|---|
| canonical | 1.000 | 0.000 | 125.000 | yes |
| balanced | 0.772 | 1.000 | 95.802 | yes |
| aggressive | 0.697 | 1.000 | 87.115 | yes |
| robust | 1.000 | 0.000 | 125.000 | yes |

Table 7: Readiness gate breakdown.

| Gate | Score | Weight |
|---|---|---|
| Core Validity Artifacts | 2.00 | 2.00 |
| Baseline Coverage/Tau Strength | 1.50 | 1.50 |
| Seed Robustness | 2.00 | 2.00 |
| Split Robustness (val vs dev_holdout) | 1.00 | 1.00 |
| Topology Signal Strength | 1.50 | 1.50 |
| Baseline Comparator Package | 1.00 | 1.00 |
| External Dataset Generalization | 0.50 | 0.50 |
| Proof Maturity | 0.50 | 0.50 |

lication risk is primarily reviewer-facing (policy conservatism optics and replay-grid granularity), not artifact consistency.

## References


David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.

Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533, 2002.

Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021.

Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371–421, 2008.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

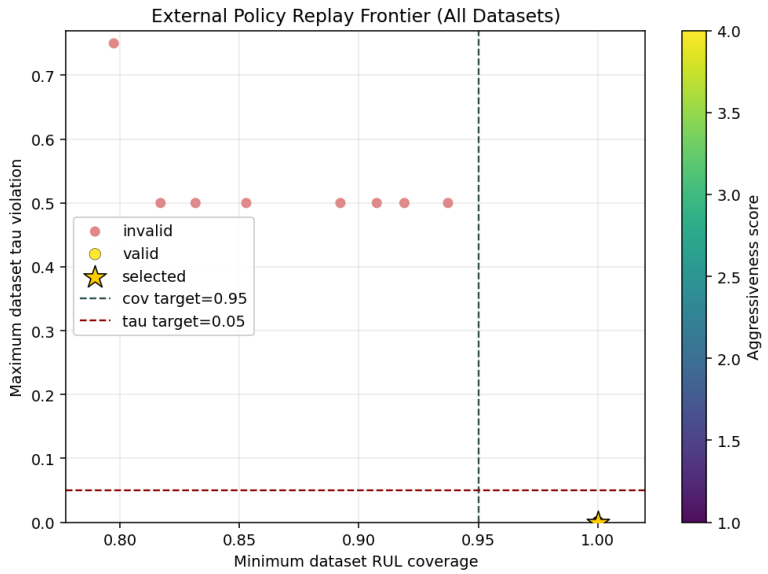


Figure 1: External replay frontier over policy settings. Vertical and horizontal dashed lines denote coverage and tau targets; star marks selected robust point.

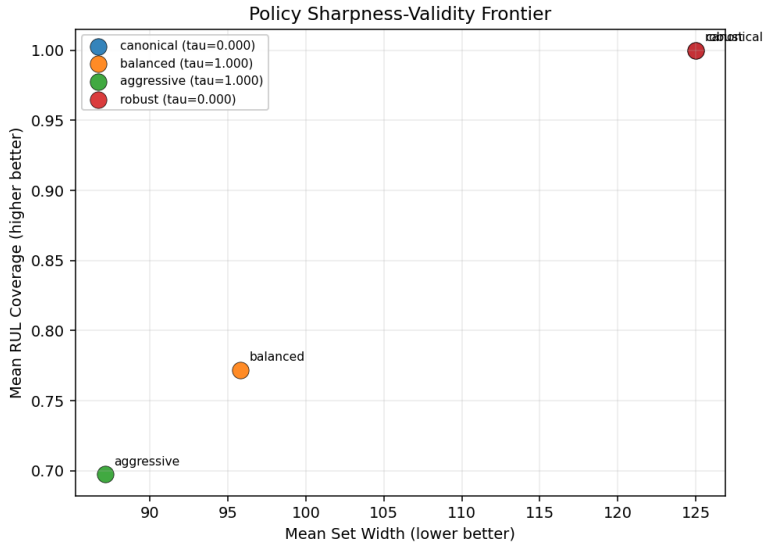


Figure 2: Policy sharpness-validity frontier (mean set width vs mean coverage; legend includes tau max).

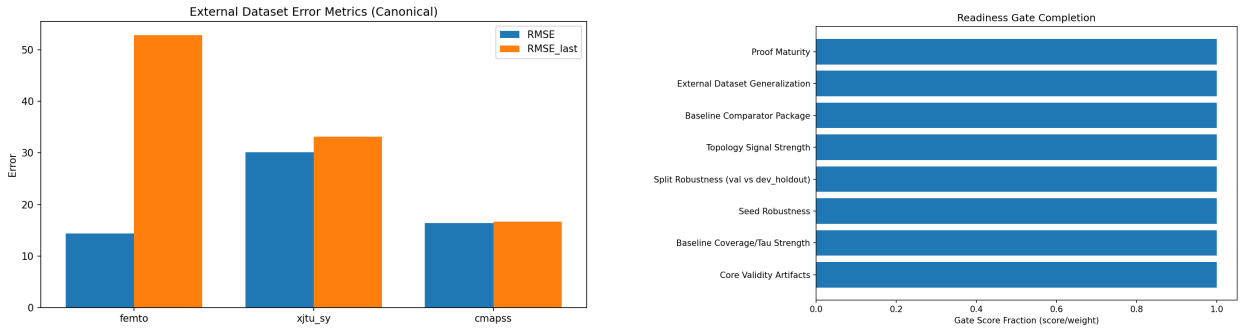


Figure 3: Canonical external errors and readiness gate completion.

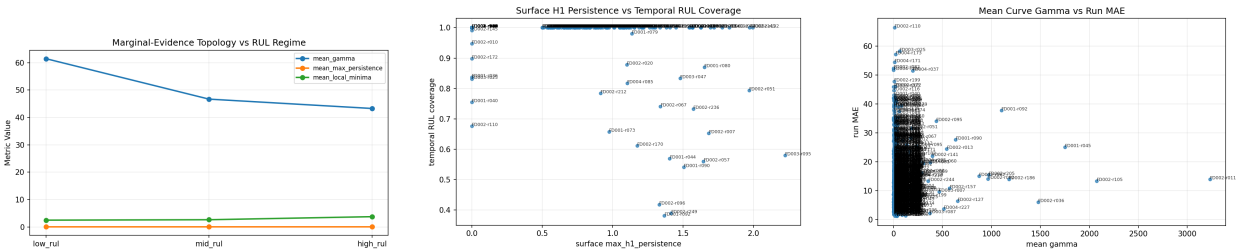


Figure 4: Topology-behavior diagnostics from canonical publication artifacts.